

# AYUSH GUPTA

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## EDUCATION

### Vellore Institute of Technology

Bhopal, M.P.

Bachelor of Technology, CSE (specialization in Health Informatics), Cumulative GPA: 8.27

2021 – 2025

**Relevant Coursework:** Database Management System, Operating Systems, Data Structures & Algorithms, Probability and Statistics, Data Science, Data Mining and Data Warehousing, Machine Learning, Artificial Intelligence in Healthcare

## SKILLS SUMMARY

**Programming Languages & Libraries:** Python, SQL, NumPy, Pandas, Matplotlib, Scikit-Learn

**Analytical Skills:** Data Cleaning, Data Modeling, Exploratory Data Analysis, Feature Engineering, Predictive Modeling

**Visualization Tools:** Tableau, Power BI, Microsoft Excel

**Databases:** MySQL, PostgreSQL, SQLite

**Cloud Platforms:** AWS, Microsoft Azure

**Developer Tools:** GitHub, Visual Studio Code, Jupyter Notebook

## PROJECTS

### HealthSage : Your Health Assistant | Python, Scikit-Learn, NumPy, Pandas, Matplotlib, SQLite, Tkinter

[GitHub](#)

- Engineered a medical diagnostic system with 96% accuracy by implementing ensemble machine learning algorithms (Decision Tree, Random Forest, Naive Bayes, KNN) to predict diseases from patient symptoms.
- Designed a user-friendly Tkinter-based GUI to analyze 95+ symptoms and diagnose 41 diseases, streamlining patient screening and reducing initial assessment time compared to manual assessment.
- Improved system reliability by incorporating cross-validation and confusion matrix analysis, enhancing prediction accuracy by 15% and reducing false positives by 30%.

### Netra Rakshak | Python, Numpy, Pandas, Matplotlib, Scikit-Learn, PyTorch, Hugging Face

[GitHub](#)

- Developed an AI-based diagnostic tool for diabetic retinopathy, achieving 96% testing accuracy by leveraging Transfer Learning with ResNet-152 on fundus scan datasets.
- Optimized detection by implementing data augmentation, normalization, ResNet-152 transfer learning, and class balancing, improving accuracy by 15% over baseline models.
- Deployed the model on Hugging Face Spaces, enabling seamless analysis of over 1,000 images, enhancing accessibility for medical professionals.

## CERTIFICATIONS

AWS Academy Graduate - AWS Academy Cloud Foundations

[Certificate](#)

The Economics of Health Care Delivery, Wharton School, University of Pennsylvania (Coursera)

[Certificate](#)

## ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- National Finalist**, Health-Tech and AI Hackathon (SPIRIT'25), IIT - BHU
- National Finalist**, Excel at Excel: Escape Room Competition (Backwaters'25), IIM Kozhikode
- Event Management Co-Lead – The Bioengineering Club, VIT Bhopal:** Directed planning and execution of flagship events including BioExpo'23 and RABVIT'24, attracting 1,000+ participants and boosted engagement by 20%. Hosted renowned speakers like Dr. Gobardhan Das (Director, IISER Bhopal) and Dr. Amitav Mishra (Professor, IGNOU).
- Student Volunteer, Health Hackathon – Improving Health Through Accessible and Empowering Innovation, VIT Bhopal University & Johns Hopkins University:** Assisted in organizing and coordinating the hackathon by providing technical support and facilitating mentor-team interactions for this healthcare innovation-focused event.
- Fundraising Volunteer, HelpAge India Foundation:** Raised Rs.20,000 for elderly welfare programs through targeted community outreach campaigns, increasing donor engagement by 15% compared to previous years. Honored with the distinguished "Star Volunteer" badge and a certificate of recognition personally signed by former members of the Indian Men's Cricket Team.